# Report 1: Preliminary Analysis of Noun Memorability and Syntactic Structure in Sentence Recognition

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## 1 Introduction and Background

Early transformational accounts of sentence memory — critically examined by both Bregman and Strasberg (1968) and James et al. (1973) — held that speakers encode sentences by reducing them to a canonical "kernel": a simple, active, affirmative, declarative form, supplemented by syntactic tags indicating the transformations required to restore the original surface structure. Under this *kernel-plus-tag* model, the passive voice, negation, and interrogative form are each stored as single, independently forgettable memory-trace components. A clear empirical consequence follows: if syntactic features are genuinely independent and all-or-none, then a participant who fails to identify the correct syntactic form on a first attempt should perform at chance on a second attempt, because the relevant trace component would simply be absent (Bregman and Strasberg, 1968).

Bregman and Strasberg (1968) tested this prediction directly. They presented 20 five-word sentences — each instantiated in active, passive, negative, and question forms — to 60 undergraduates in a forced-choice recognition procedure. After participants committed to a first-guess syntactic label, Bregman and Strasberg examined whether second guesses on errors still carried information. If syntactic features were all-or-none traces, they should not. Instead, second-guess accuracy was reliably above chance across both replications ($\chi^2(4) = 20.8,\ p < .001$). A shift analysis revealed a consistent *active–passive subcategory*: participants who confused active with passive sentences did so systematically, while negative and question confusions arose largely as a forced-choice artefact. Post-experimental interview data confirmed that participants reconstructed the active–passive distinction from retained semantic information — action imagery, truth value, affective response, and noun salience — rather than from any stored syntactic symbol

(Bregman and Strasberg, 1968). Bregman and Strasberg therefore concluded that once a sentence has been comprehended, the properties of the transmission code need not be preserved in the memory trace; what enters long-term memory refers to the *semantic* properties of the message, not the surface form in which the message was delivered.

This conclusion carries a specific implication: the semantic substance retained from a sentence — its "gist" — should be largely decoupled from its verbatim wording. Begg (1971) examined this dissociation directly in a continuous recognition paradigm. Using two parallel lists of 502 sentences drawn from novels and rated for imagery, Begg tested eight participants on meaning judgments ("Have you seen a sentence with this meaning before?") and wording judgments ("Are these the exact words you saw?") across lags of 2 to 64 intervening sentences. Meaning recognition declined monotonically and substantially with lag ($r = -.996$, $p < .001$), whereas wording recognition was entirely uncorrelated with lag ($r = -.107$, *n.s.*) and remained above chance at approximately 0.62 correct, a level consistent with wording memory relying on reconstruction rather than on an intact trace. Furthermore, meaning and wording judgments were stochastically independent: the probability of a jointly correct judgment approximated the product of the marginal probabilities ($P_{cM} \cdot P_{cW} = .778$ vs. observed $P_{cM\&W} = .790$), indicating that the two systems draw on distinct memory representations (Begg, 1971). Begg interpreted these findings as support for a dual-trace model in which concrete sentence meaning is stored as a nonverbal spatial image while specific words are reconstructed from that image at retrieval rather than preserved as a sequential verbal string. Crucially, meaning recognition was no less accurate for paraphrase probes than for identical-wording probes, confirming that syntactic and lexical cues provide no facilitation when retrieving the stored meaning image (Begg, 1971).

If both syntactic tags and verbatim word sequences are unreliably retained, the question shifts to what drives the reconstructive process when the gist image is retrieved. James et al. (1973) identified two systematic biases. First, spontaneous speech production favours active constructions; because recall is itself a constructive act governed by the same production preferences, participants disproportionately reconstruct passive sentences as active. Second, and more directly relevant to the present study, speakers tend to initiate a recalled sentence with the most salient noun in the stored semantic representation — operationalised as imagery value *(I-value)* — regardless of whether that noun originally occupied the surface subject or object position. James and colleagues instantiated this manipulation in a 4 (noun imageability: HH, HL, LH, LL) $\times$ 2 (voice: active, passive) free-recall design, selecting high-imageability nouns (ratings $> 5.90$) and low-imageability nouns (ratings $< 3.50$) on a 7-point scale. The voice-error probability data (James et al., 1973, Table 2) were highest for LH sentences (0.245 active; 0.333 passive) and lowest for HL sentences (0.029 active; 0.049 passive), with HH and LL at

intermediate and approximately equal rates — precisely the pattern predicted by a reconstructive account in which the high-imageability noun captures the sentence's "theme" and is preferentially placed in subject position at recall (James et al., 1973). This ordering was inconsistent with all five storage hypotheses evaluated — including the kernel-plus-tag model — without supplementary reconstructive assumptions. Overall sentence recall also scaled monotonically with imagery load: HH > HL ≈ LH > LL (James et al., 1973, Table 1).

Taken together, these three lines of work converge on a framework in which sentence memory is driven primarily by the semantic and imagistic properties of constituent words, while verbatim syntactic form occupies a secondary, independently stored representation that decays more rapidly and is reconstructed rather than retrieved. The present experiment builds directly on this framework by adapting the noun-imageability manipulation of James et al. (1973) to a *recognition* rather than recall paradigm, using a continuous presentation procedure modelled on Begg (1971). Whereas James and colleagues measured voice-error probability in free recall, we assess *corrected* hit rates for sentence meaning recognition (Immediate Recognition, IR) and separately measure participants' ability to discriminate the grammatical voice of the encoded sentence (Wording Recognition, WR). This dual-response design mirrors Begg's separation of meaning and wording judgments, enabling us to test whether noun imageability selectively affects semantic gist encoding (IR), surface-form retention (WR), or both. If the reconstructive framework is correct, imageability should affect IR — because the richness of the image trace governs whether gist is recognised upon second presentation — while WR should remain largely insensitive to noun imageability but may show a directional advantage for active over passive voice, consistent with the production bias documented by James et al. (1973).

# 2 Methods

## 2.1 Experimental Design and Stimuli

This study employed a $4 \times 2$ fully within-participants design, with Noun Condition (HH, HL, LH, LL) and Grammatical Voice (Active, Passive) as the two factors. Simple Subject-Verb-Object (S-V-O) sentences were generated by systematically varying the imageability of the subject and object nouns. High- and low-imageability nouns were drawn from established word imageability norms following the same operationalisation used by James et al. (1973), while verbs were selected from a fixed set of medium-concreteness action verbs to minimise variability across conditions. Automated fluency checks and semantic-coherence filters removed ungrammatical or implausible sentences, followed by manual screening.

This procedure yielded four noun conditions:

- **HH:** High-imageability subject, High-imageability object.
- **HL:** High-imageability subject, Low-imageability object.
- **LH:** Low-imageability subject, High-imageability object.
- **LL:** Low-imageability subject, Low-imageability object.

From the resulting pool, 200 active-voice sentences were selected (50 per noun condition). Each sentence was then converted into a passive-voice equivalent, producing paired active and passive variants. This resulted in 400 target sentences across eight experimental conditions (4 noun conditions $\times$ 2 voices). Sentences were kept short ($\leq$ 8 words) and excluded proper nouns to inhibit extraneous mnemonic associations. An additional set of high-frequency filler sentences (prefix `HF`) was interwoven throughout the stream to provide false alarm opportunities and mitigate list-learning effects, following the continuous recognition approach of Begg (1971).

## 2.2 Participants and Procedure

A priori power analysis indicated that 334 valid participants would be required to achieve adequate statistical power for detecting interaction effects of the anticipated magnitude. Ultimately, 114 online participants began the experiment, providing a preliminary dataset.

Participants completed a continuous recognition task comprising three blocks separated by rest phases to limit fatigue. Each participant was exposed to 48 target sentences (16 per block), balanced across the 8 experimental conditions. Each target sentence appeared twice: once as an *encoding* trial (first presentation) and once as a *probe* trial (second presentation, the recognition test). Stimulus IDs encoded experimental conditions: the prefix denoted noun condition (`HH`, `HVL` $\rightarrow$ HL, `LVH` $\rightarrow$ LH, `LVL` $\rightarrow$ LL) and the trailing character denoted voice (`A` = active, `P` = passive).

Upon seeing a probe, participants provided two sequential responses:

1. **Immediate Recognition (IR):** Participants pressed the Spacebar if they recognised the sentence's meaning as a repeat from earlier in the session. This corresponds to Begg's (1971) meaning judgment.
2. **Wording Recognition (WR):** Immediately following an IR press, participants made a forced-choice "Yes/No" judgment as to whether the probe matched the exact grammatical voice (active vs. passive) of its initial presentation. This corresponds to Begg's (1971) wording judgment.

## 2.3 Data Cleaning and Exclusion Criteria

Raw event logs were processed through a dedicated R pipeline. Practice-phase rows, inter-stimulus gap-time metadata events, and rest-phase markers were removed. A sequential `block_id` (1–3) was assigned per participant based on rest-phase boundaries.

An objective block-level validity criterion was applied, based on performance on embedded validation sentences. Three event types were used: *correct validation* (IR pressed on a validation repeat), *wrong IR press* (IR pressed on a non-repeat sentence), and *missed validation* (no IR press on a validation repeat). A block was retained only if:

$$\text{Correct Validations} > \frac{\text{Wrong IR Presses}}{2} + \text{Missed Validations}$$

Blocks failing this criterion were excluded from all downstream analyses. This block-level filter retained 329 of a possible 336 participant-blocks (112 of 114 participants; 97.9% pass rate). Block-level participant counts after exclusion were: Block 1 ($n = 108$), Block 2 ($n = 109$), Block 3 ($n = 112$).

## 2.4 Computed Measures and Analytical Approach

**IR Corrected Recognition.** To isolate discrimination sensitivity from individual response bias, the primary outcome was the corrected recognition score:

$$\text{Corrected IR} = \text{Hit Rate} - \text{False Alarm (FA) Rate}$$

*Hit rate* was calculated per participant per noun condition (collapsed across voice for the primary Kruskal-Wallis test; by condition × voice for the 8-cell analyses): the proportion of target probe trials on which the participant pressed the Spacebar. Missed probes counted as 0. *False alarm rate* was calculated globally per participant as the proportion of HF-filler first showings that received an erroneous IR press; validation sentences were excluded from the FA denominator because they serve a different experimental function. The global FA rate was then subtracted uniformly from all condition-level hit rates for that participant. This approach isolates true discrimination from the liberal-versus-conservative response bias documented in Figure 2, paralleling signal-detection corrections while avoiding the normality assumption required by $d'$, which is inappropriate for the bounded $[-1, 1]$ range of these scores.

**WR Accuracy.** For each target probe on which an IR press was made, a wording judgment was recorded (`wr_accuracy = 1` correct; 0 incorrect). WR accuracy was computed as the proportion of correct wording judgments per participant per condition, mirroring the wording judgment measure in Begg (1971).

**IR Reaction Time.** Reaction time (RT) was extracted from the `ir_reaction_time_ms` column on IR-pressed events for target probes. Implausible RTs ($>$ 10,000 ms) were excluded.

**Statistical Tests.** All analyses aggregated scores to the participant level first, to avoid pseudoreplication. Differences across the four noun conditions were assessed with Kruskal-Wallis tests, chosen because bounded accuracy scores violate the normality assumptions of parametric ANOVA. Effect size was quantified using $\varepsilon^2 = (H-k+1)/(n-k)$, where $H$ is the Kruskal-Wallis statistic, $k$ is the number of groups, and $n$ is the number of participants. Significant omnibus tests were followed by pairwise Wilcoxon rank-sum tests with Holm–Bonferroni correction. Voice effects (Active vs. Passive) were assessed with paired Wilcoxon signed-rank tests. Above-chance WR performance was assessed with one-sample Wilcoxon tests against $\mu = 0.5$. Monotonic associations were quantified using Spearman's $\rho$. All analyses were conducted in R (version $\geq$ 4.3) using `dplyr`, `ggplot2`, `tidyr`, `stringr`, and `scales`. Figures were saved at 300 DPI.

# 3 Analysis and Results

## 3.1 Descriptive Overview and Response Bias

Across the 5,264 valid target probe trials, the overall mean hit rate was 0.832 and the mean false alarm rate was 0.160 (range: 0.000–0.427). After correction, the overall mean corrected IR was $M = 0.673$ ($Mdn = 0.708$, range: $-0.141$–1.000).

As shown in Figure 2, there is a significant positive correlation between hit rate and false alarm rate at the participant level (Spearman $\rho = .208$, $p = .028$), confirming that more liberal responders accumulate both more hits and more false alarms simultaneously. This response bias motivates the corrected recognition measure as the primary dependent variable. Response time distributions (Figure 1) display the characteristic right-skewed shape expected for memory retrieval latencies, with the bulk of responses concentrated between 800 and 2,500 ms.

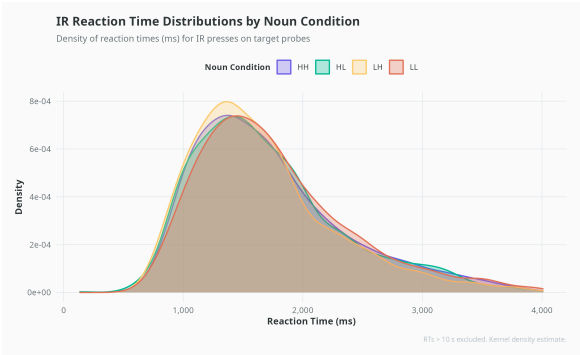


Figure 1: Density distributions of Immediate Recognition reaction times (ms) by noun condition. All four conditions show the characteristic right-skewed latency profile of memory retrieval; RTs $>$ 10,000 ms excluded.

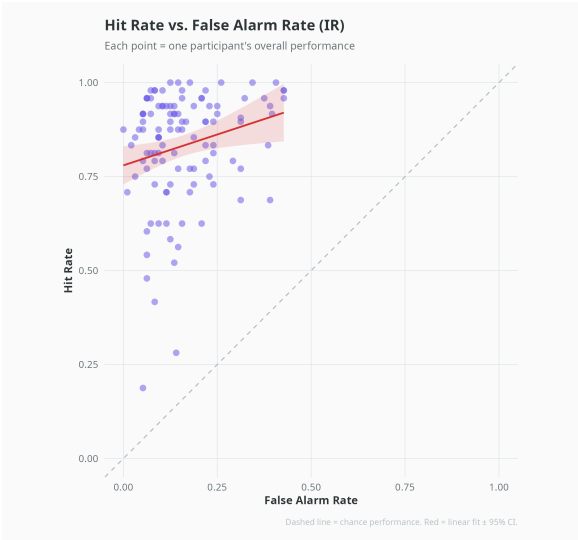


Figure 2: Per-participant hit rate against false alarm rate. The positive linear trend (Spearman $\rho = .208$, $p = .028$) confirms systematic response bias, motivating the corrected recognition score. The dashed diagonal represents chance (hit = FA).

The per-participant heatmap (Figure 3) illustrates substantial between-participant variability in baseline performance. The consistent colour shift from HH/HL/LH (warmer greens) to LL (cooler reds) within individual rows confirms that the noun-condition effect is not driven by a small subset of participants.

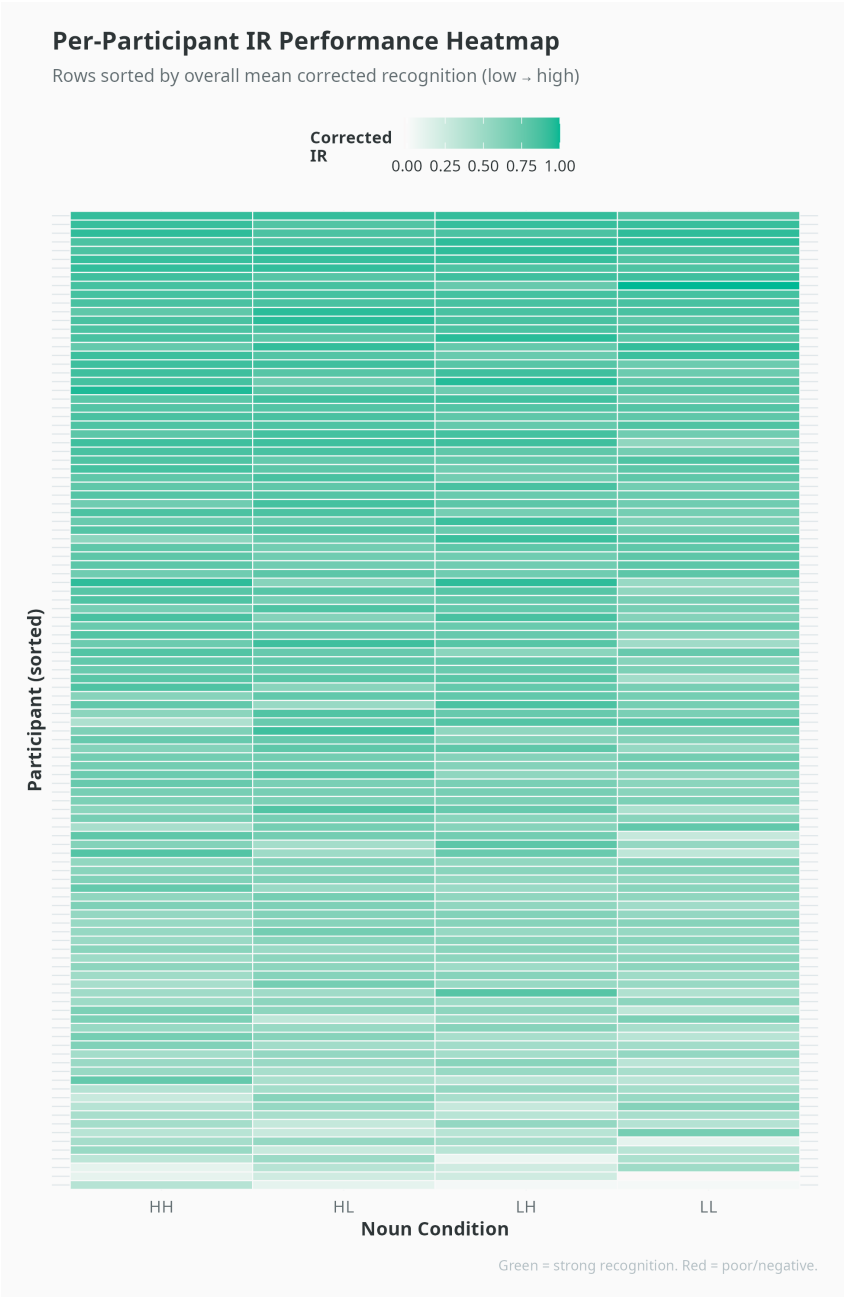


Figure 3: Per-participant corrected IR heatmap. Rows are sorted by overall mean corrected recognition (low to high). The rightward shift to cooler colours in the LL column is visible within most participants' rows, indicating a robust within-subject effect.

## 3.2 Main Effect of Noun Condition on Immediate Recognition

We predicted that noun imageability would positively influence sentence recognition, following the recall pattern documented by James et al. (1973). Table 1 summarises corrected IR scores collapsed across grammatical voice.

Table 1: Corrected Immediate Recognition (IR) Scores by Noun Condition ($n = 112$)

| Condition | $M$ | $SD$ | $Mdn$ | Range |
|---|---|---|---|---|
| HH | 0.689 | 0.185 | 0.740 | −0.141 – 1.000 |
| HL | 0.691 | 0.177 | 0.708 | −0.052 – 1.000 |
| LH | 0.679 | 0.188 | 0.729 | −0.083 – 0.979 |
| LL | 0.632 | 0.189 | 0.656 | −0.141 – 1.000 |

A Kruskal-Wallis test revealed a statistically significant main effect of noun condition on corrected IR, $H(3) = 8.85$, $p = .031$, $\varepsilon^2 = .054$ (small-to-medium effect). Holm-corrected pairwise Wilcoxon tests indicated that the LL condition showed marginally lower corrected recognition than HL ($p = .058$) and HH ($p = .070$), and a non-significant trend relative to LH ($p = .130$). All pairwise comparisons among HH, HL, and LH were entirely non-significant (all $p = 1.00$; Figure 4).

This pattern is consistent with a *threshold effect*: the presence of at least one high-imageability noun appears sufficient to sustain near-maximal recognition, with no incremental benefit from a second high-imageability noun. Notably, this contrasts with the graded recall advantage documented by James et al. (1973), who found HH > HL ≈ LH > LL in free-recall accuracy. The absence of a graded pattern here may reflect a lower encoding threshold in recognition compared to the more demanding explicit retrieval required in recall — a distinction worth examining with the full planned sample in Report 2.

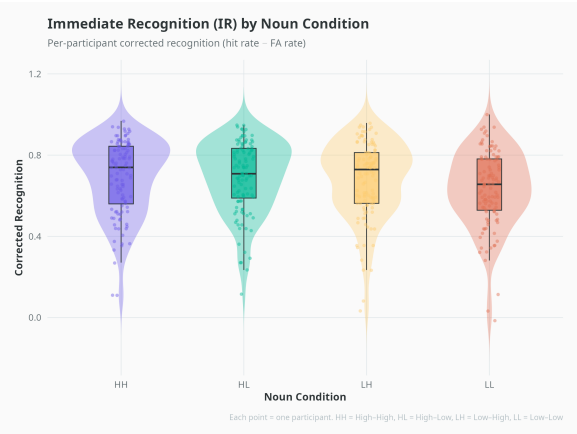


Figure 4: Main effect of noun condition on corrected IR. Violin contours show the full participant-level distribution; boxes show interquartile range; points show individual participants. The LL condition is the only one with a visibly depressed distribution.

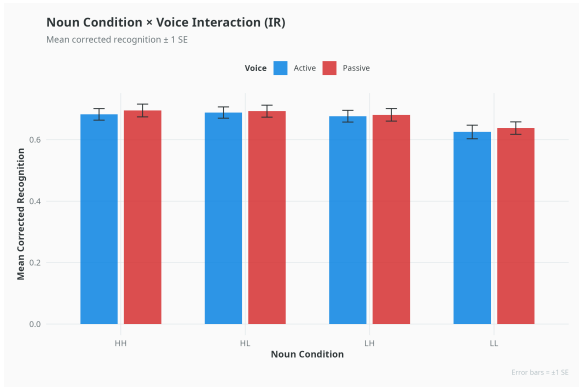


Figure 5: Mean corrected IR (±1 SE) across the 8 noun-condition × voice cells. The LL disadvantage replicates within both active and passive voice, with no evidence of an interaction.

## 3.3 Noun Condition Does Not Affect Wording Recognition

In contrast to the IR result, a Kruskal-Wallis test found no significant effect of noun condition on WR accuracy, $H(3) = 1.98$, $p = .576$ (Figure 6). Mean WR accuracy by condition is presented in Table 2. Despite the large spread in IR scores across conditions, participants were equally accurate at discriminating grammatical voice regardless of noun imageability. This null result — for WR but not IR — supports the prediction derived from Begg's (1971) dual-trace model: noun imageability specifically influences the richness of the semantic gist image, while the representation underlying voice discrimination operates through independent mechanisms and is insensitive to the imagistic quality of the constituent words.

Table 2: Wording Recognition (WR) Accuracy by Noun Condition ($n = 112$)

| Condition | $M$ | $SD$ | Note |
|---|---|---|---|
| HH | 0.748 | 0.151 | above chance |
| HL | 0.747 | 0.145 | above chance |
| LH | 0.762 | 0.160 | above chance |
| LL | 0.730 | 0.173 | above chance |

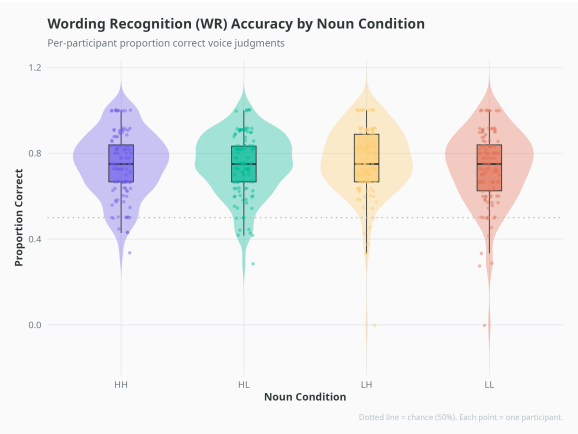


Figure 6: WR accuracy by noun condition. All four distributions are clearly above the 0.50 chance line (dotted), confirming that participants retained surface-form information. No significant condition differences were found ($H(3) = 1.98$, $p = .576$).

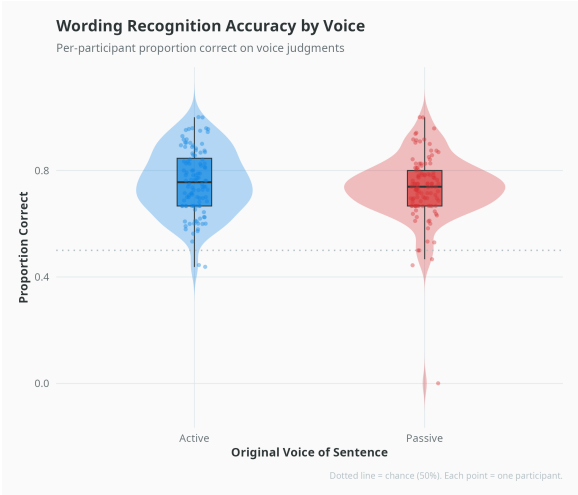


Figure 7: WR accuracy by grammatical voice. Both voices are well above chance. The numerical advantage for Active ($M = 0.757$) over Passive ($M = 0.735$) did not reach significance ($V = 3279$, $p = .113$).

## 3.4 Voice Effects on IR and WR

**Effect of voice on IR.** A paired Wilcoxon signed-rank test found no significant main effect of grammatical voice on corrected IR: $V = 1704$, $p = .374$ (Active $M = 0.668$, $SD = 0.166$; Passive $M = 0.677$, $SD = 0.175$; Figures 5 and 9). Whether a sentence was encoded in active or passive voice had no measurable effect on how well its meaning was recognised — a direct replication of Begg's (1971) finding that meaning recognition is equally accurate for paraphrase and identical-wording probes. The 8-cell means in Table 3 confirm that the noun-condition threshold effect replicates consistently within both voice conditions, with no evidence of a differential interaction.

Table 3: Mean Corrected IR by Noun Condition × Voice ($n = 112$ per cell)

| Condition | Active $M$ | Active $SD$ | Passive $M$ | Passive $SD$ |
|---|---|---|---|---|
| HH | 0.683 | 0.199 | 0.695 | 0.220 |
| HL | 0.689 | 0.195 | 0.693 | 0.206 |
| LH | 0.677 | 0.202 | 0.681 | 0.216 |
| LL | 0.625 | 0.234 | 0.638 | 0.215 |

**Effect of voice on WR.** WR accuracy was numerically higher for active-voice sentences ($M = 0.757$, $SD = 0.120$) than passive-voice sentences ($M = 0.735$, $SD = 0.127$), but this difference did not reach statistical significance: $V = 3279$, $p = .113$ (Figure 7). Critically, one-sample Wilcoxon tests confirmed that both voices were reliably above chance: Active $V = 6322$, $p < .001$; Passive $V = 5990$, $p < .001$. Participants could accurately detect voice mismatches regardless of the original grammatical form — consistent with Bregman and Strasberg's (1968) finding that above-chance second-guess accuracy for the active–passive distinction persists even after the verbatim trace has partially degraded. The non-significant trend favouring active-voice WR is directionally consistent with the active-construction bias documented by James et al. (1973) and merits re-examination with the full sample in Report 2.

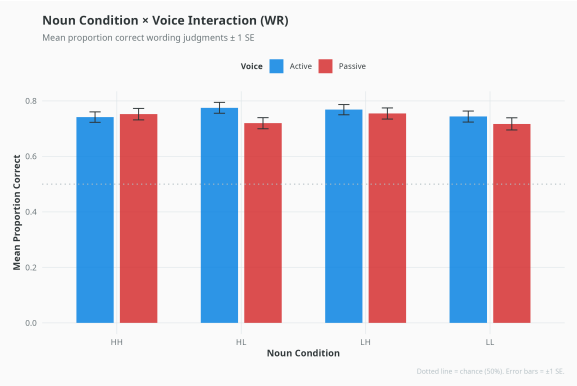


Figure 8: Mean WR accuracy (±1 SE) across 8 noun-condition × voice cells. Performance is uniformly high across all cells with no discernible interaction, consistent with the null noun-condition effect on WR.

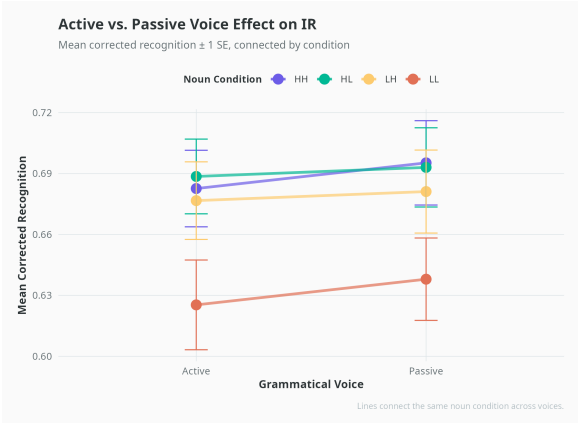


Figure 9: Voice slope graph for IR. Near-parallel lines indicate that the absence of a voice effect generalises across all four noun conditions, with no crossover interaction.

## 3.5 Relationship Between Meaning and Wording Memory

Although IR and WR are predicted to operate on distinct memory representations (Begg, 1971), they were significantly and positively correlated at the participant level (Spearman $\rho = .248$, $p < .001$; Figure 10). Participants who more reliably recognised sentence meaning also showed superior voice discrimination. The positive sign of this association is consistent with Begg's (1971) observation that the product rule for stochastic independence ($P_{cM} \cdot P_{cW} \approx P_{cM\&W}$) holds only approximately: the two systems are largely but not entirely independent. The modest magnitude ($\rho = .25$) confirms that the two memory systems are not redundant, in line with the partially dissociable dual-trace account.

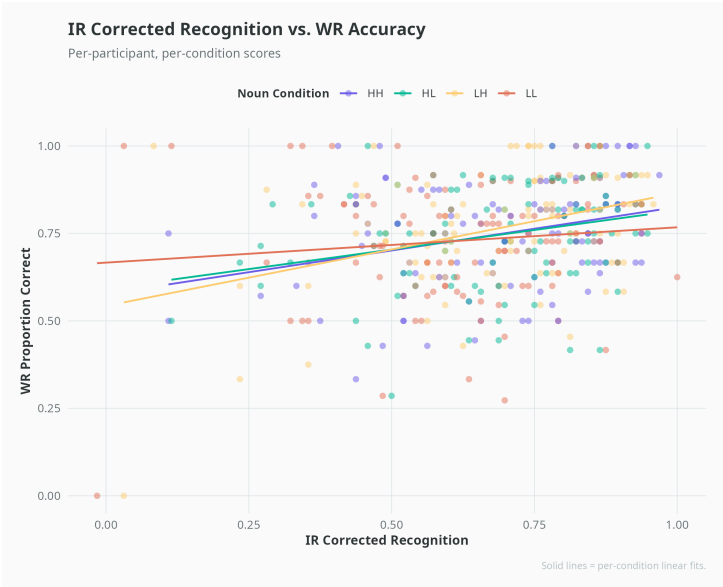


Figure 10: Scatterplot of per-participant, per-condition corrected IR against WR accuracy (Spearman $\rho = .248$, $p < .001$). Separate linear fits are shown for each noun condition. The positive relationship suggests co-occurring encoding strengths; the modest effect size indicates partially dissociable representations, consistent with Begg (1971).

## 3.6 Reaction Time as a Supplementary Processing Index

Mean IR reaction times by noun condition were: HH $M = 1{,}725$ ms ($SD = 319$), HL $M = 1{,}729$ ms ($SD = 365$), LH $M = 1{,}677$ ms ($SD = 335$), and LL $M = 1{,}762$ ms ($SD = 336$). A Kruskal-Wallis test found no significant effect, $H(3) = 4.44$, $p = .218$ (Figure 1). Although LL showed numerically the slowest median responses and LH the fastest, these differences did not reach significance. The similarity of RT distributions across conditions suggests that the noun-condition effect on accuracy is not accompanied by a corresponding speed–accuracy trade-off.

## 3.7 Block Effects: Proactive Interference

Mean corrected IR declined numerically across blocks (Block 1: $M = 0.699$, $SD = 0.151$, $n = 108$; Block 2: $M = 0.679$, $SD = 0.195$, $n = 109$; Block 3: $M = 0.650$, $SD = 0.191$, $n = 112$). A Kruskal-Wallis test found this trend to be statistically non-significant: $H(2) = 2.77$, $p = .250$ (Figure 11). The ∼2–3 percentage point decline per block is consistent with mild proactive interference — analogous to the lag effect on meaning recognition reported by Begg (1971) — but does not approach significance at this sample size and is unlikely to have materially distorted the noun-condition comparisons.

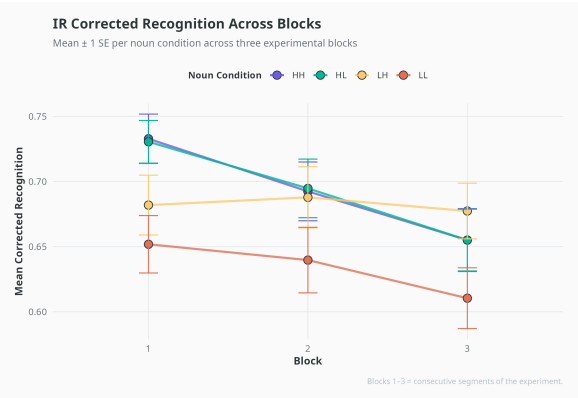


Figure 11: Mean corrected IR (±1 SE) by noun condition across Blocks 1–3. The downward trend is consistent with mild proactive interference but does not reach significance ($H(2) = 2.77$, $p = .250$). The LL condition remains consistently below the other three throughout.

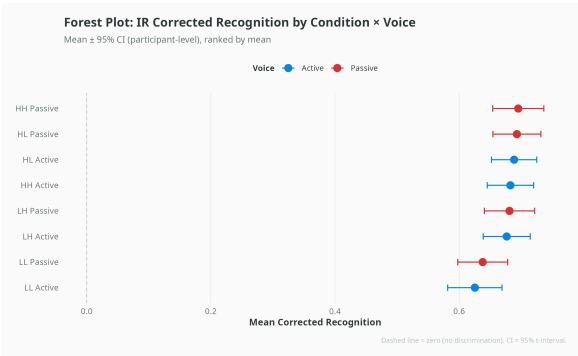


Figure 12: Forest plot of mean corrected IR (±95% CI) for all 8 condition × voice cells, ranked by mean. The LL conditions cluster at the lower end; all 95% CIs exclude zero, confirming above-chance recognition in every cell.

## 3.8 Summary of Statistical Tests

The forest plot (Figure 12) synthesises all eight condition × voice cells. The LL Active and LL Passive conditions cluster at the lower end; HH, HL, and LH conditions form an elevated cluster between approximately $M = 0.67$ and $M = 0.70$ with no systematic active–passive stratification visible, corroborating the null voice effect on IR. All 95% confidence intervals exclude zero, confirming above-chance recognition in every experimental condition. Table 4 summarises all primary tests.

Table 4: Summary of Primary Statistical Tests

| Test | Statistic | *df* | *p* | Significant |
|---|---|---|---|---|
| KW: IR × Noun Condition | $H = 8.85$ | 3 | .031 | Yes |
| KW: WR × Noun Condition | $H = 1.98$ | 3 | .576 | |
| KW: IR × Block | $H = 2.77$ | 2 | .250 | |
| KW: RT × Noun Condition | $H = 4.44$ | 3 | .218 | |
| WSR: IR Active vs. Passive | $V = 1704$ | – | .374 | |
| WSR: WR Active vs. Passive | $V = 3279$ | – | .113 | |
| WR vs. chance — Active (one-sample) | $V = 6322$ | – | $< .001$ | Yes |
| WR vs. chance — Passive (one-sample) | $V = 5990$ | – | $< .001$ | Yes |

# 4 Conclusion and Future Directions

This preliminary analysis provides clear evidence that noun imageability influences sentence recognition, consistent with the reconstructive memory framework advanced by James et al. (1973) and Begg (1971). The significant Kruskal-Wallis effect ($H(3) = 8.85$, $p = .031$, $\varepsilon^2 = .054$) confirms that sentences containing only low-imageability nouns (LL) are recognised less reliably than those containing at least one high-imageability noun. Critically, this recognition pattern takes a *threshold* form rather than the graded one documented by James et al. (1973) in free recall, where HH > HL ≈ LH > LL. This divergence is theoretically meaningful: in recall, each noun's imageability contributed additively to reconstruction success, whereas in recognition a single high-imageability noun appears sufficient to generate a retrievable image trace. Whether this reflects a fundamental difference between recognition and recall, or simply a task-difficulty ceiling, requires the full planned sample and mixed-effects modelling to resolve.

Grammatical voice had no detectable influence on IR ($V = 1704$, $p = .374$), directly replicating Begg's (1971) finding that meaning recognition is equally accurate for identical and paraphrase probes. Wording recognition (WR), while reliably above chance for both active ($V = 6322$, $p < .001$) and passive sentences ($V = 5990$, $p < .001$) — consistent with Bregman and Strasberg's (1968) above-chance second-guess performance for the active–passive distinction — was also unaffected by noun imageability ($H(3) = 1.98$, $p = .576$). This double dissociation confirms that the mechanism supporting voice discrimination is independent of the semantic encoding strength that drives IR. The modest positive correlation between IR and WR (Spearman $\rho = .248$, $p < .001$) is consistent with Begg's (1971) approximately independent but co-varying dual-trace model.

**Next Steps for Report 2.** The present sample ($N = 114$) is substantially below the targeted $N = 334$, limiting statistical power for detecting interaction effects and precluding parametric modelling. Report 2 will address this through:

1. **Generalised Linear Mixed Models (GLMMs):** Mixed-effects logistic regression will model IR accuracy as a function of Noun Condition and Voice as fixed effects, with random intercepts and slopes for both Participant and Item. This framework accounts for by-participant and by-item variability simultaneously, substantially increasing power relative to the participant-aggregated non-parametric tests used here.

2. **Reaction Time Modelling:** Log-transformed RT will be analysed with linear mixed models to test whether noun imageability predicts recognition speed — a proxy for trace strength — independent of its effect on accuracy. A speed advantage for high-imageability conditions would provide converging evidence for the image-trace hypothesis of Begg (1971).

3. **Confirmatory Power Analysis:** Effect sizes from the present preliminary analysis (KW $\varepsilon^2 = .054$) will be used to verify that the targeted $N = 334$ provides $\geq 80\%$ power for all primary contrasts.

# References


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